

ERROR HANDLING IN COMPILERS

CSA1407-Compiler Design

Technical Presentation

Name : Sabitha. K

Reg no:192524308

Dept : AIDS

INTRODUCTION TO ERROR HANDLING

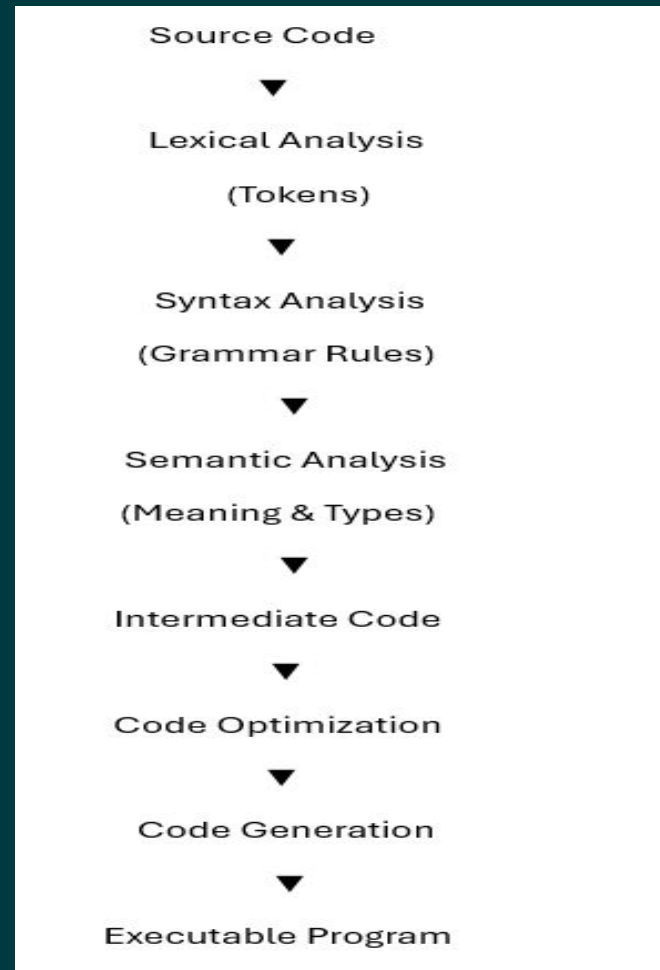
Definition:

Error handling is the process by which a compiler **detects, reports, and recovers from errors** in a program during compilation.

Objectives

- ✓ Detect programming errors.
- ✓ Display meaningful error messages.
- ✓ Continue compilation whenever possible.
- ✓ Help programmers debug Time.

COMPILER PHASES & WHERE ERROR HANDLING STARTS



PROBLEM STATEMENT AND SOLUTION

PROBLEM:

- Programmers often make errors while writing source code.
- Stopping at the first error makes debugging difficult and time-consuming.
- An efficient error handling mechanism is required to detect, report, and recover from errors.

SOLUTION:

- Detect errors during **Lexical, Syntax, and Semantic Analysis**.
- Display clear and meaningful error messages with the error location.
- Report multiple errors in a single compilation.
- Improve debugging efficiency, code quality, and compiler reliability

Types of Errors

Error Type	Example
Lexical Error	int @x;
Syntax Error	Missing ;
Semantic Error	int x = "Hello";
Runtime Error	Division by zero
Logical Error	Wrong formula or logic

ERROR HANDLING PROCESS & RECOVERY

Programmer Writes Code



Compiler Reads Code



Detect Error



Report Error



Recover from Error



Continue Compilation

Example Program with Errors

INPUT:

```
#include <stdio.h>
```

```
int main()
{
    int x = 10
    int y = "Hello";
    printf("%d", z);
}
```

OUTPUT:

Syntax Error:

Missing ';' after 10

Semantic Error:

Type mismatch

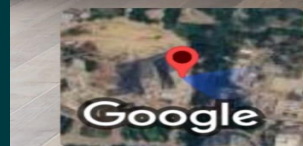
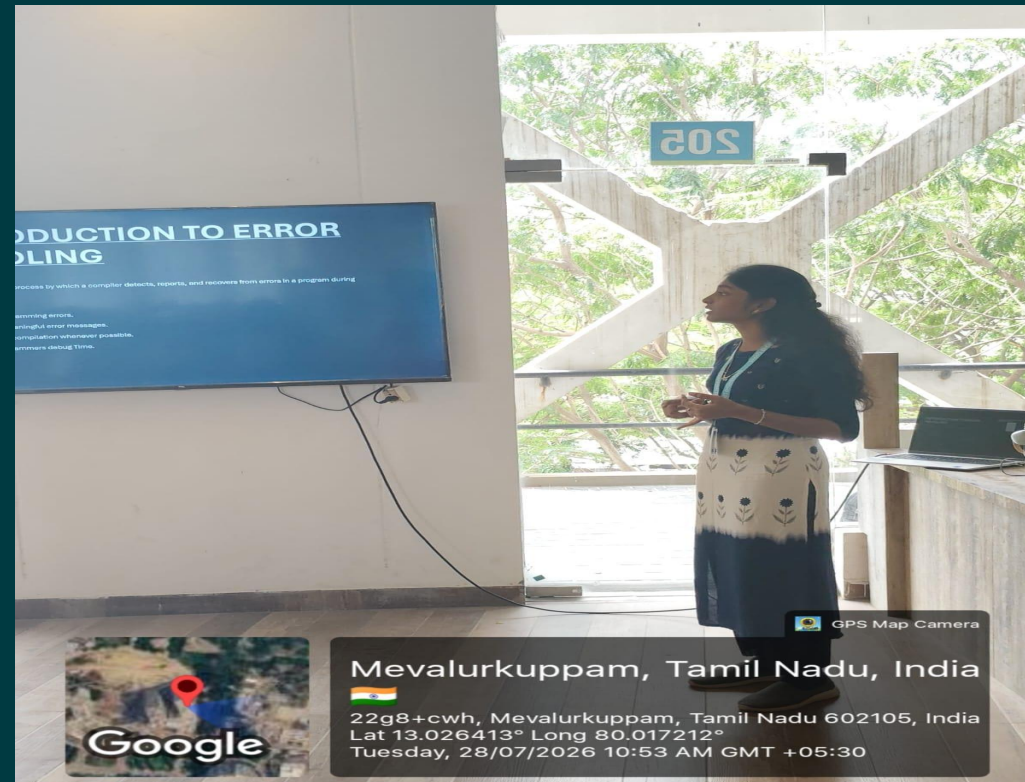
Semantic Error:

Variable 'z' not
declared

ADVANTAGES AND CONCLUSION

Benefits

- Detects errors early.
 - Saves debugging time.
 - Reports multiple errors.
 - Improves compiler reliability.
 - Enhances software quality.
 - Makes development faster and easier.
- It starts during **Lexical Analysis**.
 - Continues through **Syntax** and **Semantic Analysis**.
 - Uses recovery techniques to continue compilation.
 - Produces clear error messages to help programmers.
 - Error handling detects, reports, and recovers from programming errors.



Mevalurkuppam, Tamil Nadu, India
22g8+cwh, Mevalurkuppam, Tamil Nadu 602105, India
Lat 13.026413° Long 80.017212°
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THANK YOU